

Rohan Kumar

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PROFESSIONAL SUMMARY

Senior Data Modeler and quant analyst with 8+ years of experience designing conceptual, logical, and physical data models for financial services data platforms, warehouses, and analytical systems. Proven expertise in ER modeling, dimensional modeling (star and snowflake schemas), data dictionary creation, metadata management, and translating complex business requirements into scalable data structures. Hands-on proficiency in SQL, PostgreSQL, Azure Databricks, PySpark, SAS ETL, and big data platforms — with deep experience in data quality, data governance, ETL/ELT pipeline development, and cross-functional collaboration with data engineers, DBAs, and business analysts.

TECHNICAL SKILLS

Data Modeling	Conceptual, Logical & Physical Data Models, ER Modeling, Dimensional Modeling, Star Schema, Snowflake Schema, Entity-Relationship Design, Data Flow Diagrams, Normalization
Database Systems	PostgreSQL (primary), SQL Server, Oracle (familiar), Snowflake (familiar), MongoDB, NoSQL — query optimization, index design, schema management
Data Warehousing	Data Warehouse Architecture, Star & Snowflake Schema Design, Data Lakehouse (Azure Databricks / Delta Lake), Fact & Dimension Tables, Slowly Changing Dimensions
ETL / ELT	ETL/ELT Pipeline Development, Azure Databricks, PySpark, Apache Spark, SAS ETL, Azure Data Factory, Data Ingestion, Transformation Logic, Pipeline Optimization
Big Data Platforms	Apache Hadoop (familiar), Apache Spark / PySpark, Azure Databricks, GCP, Azure — large-scale financial dataset processing, distributed data management
SQL & Programming	SQL (advanced — joins, CTEs, window functions, query optimization), Python, SAS, R, kdb+/q — GitHub/Git, Visual Studio Code, Linux/Unix
Data Governance	Data Dictionaries, Metadata Management, Data Quality Standards, Data Lineage, Compliance (SR 11-7, CECL, SOX, SEC, FINRA), Confluence, JIRA
Modeling Tools	Azure Databricks Schema Design, Confluence Data Documentation, GitHub Version Control — ERwin/ER/Studio (familiar), PowerDesigner (familiar)

PROFESSIONAL EXPERIENCE

Madison-Davis / SMBC Compliance

Jersey City, NJ

Senior Data Modeler — Compliance Analytics Platform

Jul 2025 – Dec 2025

- Designed and maintained conceptual, logical, and physical data models for SMBC's Market Surveillance analytics platform; defined entities, relationships, attributes, and data flows for Fixed Income and FX surveillance datasets — ensuring data structures were accurate, consistent, and scalable across multiple business units and environments.
- Translated complex business requirements from Compliance, Technology, and Surveillance stakeholders into well-structured data models; created and maintained data dictionaries and metadata documentation in Confluence — establishing clear definitions for all data entities, attributes, and relationships used in downstream analytics.
- Designed and built ETL/ELT data pipelines on Azure Databricks (Python/SQL/PySpark) to ingest, transform, and load large-scale structured and unstructured financial datasets; implemented star schema-equivalent dimensional models supporting time-series analytics and aggregated performance reporting.

- Ensured data quality, consistency, and governance standards across all model inputs; defined data quality rules, implemented reconciliation workflows, and maintained data lineage documentation aligned with SEC, FINRA, and SR 11-7 compliance requirements.
- Optimized database and pipeline performance through proper indexing, partitioning, and query optimization strategies; reduced data processing time by 20–30% through schema redesign and pipeline tuning on Azure Databricks Delta Lake.
- Maintained comprehensive data model documentation, version control via GitHub, and deployment guides in Confluence; collaborated with data engineers, analysts, and risk technology teams to ensure data models supported both operational and analytical use cases.

Independent

Independent Consulting, Professional Development & Continued Education Jan 2024 – Jun 2025

- Provided independent data modeling and analytics consulting for financial institutions; deepened expertise in dimensional modeling, data warehouse design, and Snowflake data platform architecture; pursued continued education including MS Computer Science coursework with focus on database systems, data engineering, and cloud data platforms.

Deutsche Bank

Assistant Vice President, Senior Data Modeler & Analytics Engineer

New York, NY

Aug 2022 – Dec 2023

- Designed and maintained logical and physical data models for Deutsche Bank's CCAR stress testing and PPNR risk analytics platform supporting \$5B+ in corporate banking exposures, defined entities, relationships, attributes, and data flows across Corporate Trust, Institutional Cash Management, and Corporate Cash Management data domains.
- Translated complex CCAR and IRRBB business requirements into structured data models; created and maintained data dictionaries and metadata documentation for all risk model inputs, calculated fields, and output tables — achieving a 15% improvement in data accuracy and ensuring compliance with Deutsche Bank data governance standards.
- Built and maintained ETL data pipelines in Python and R to ingest macroeconomic time series, balance sheet, and exposure datasets; designed dimensional data models supporting CCAR scenario analysis, PPNR forecasting, and IRRBB sensitivity reporting.
- Ensured data quality and consistency across all model inputs; implemented data reconciliation checks, lineage tracking, and governance controls aligned with SR 11-7 and Federal Reserve regulatory requirements — supporting CCAR 2023 regulatory submission.
- Optimized SQL query performance and data pipeline throughput; applied proper indexing, partitioning, and schema normalization strategies to improve end-to-end data processing efficiency — driving a 30% improvement in cross-functional workflow performance.
- Maintained version-controlled data model documentation and deployment guides, mentored junior data analysts on data modeling best practices, SQL optimization, and metadata management standards.

Primesoft Consulting Services (Santander)

Data Modeler — Credit Risk Analytics, Contract

Jersey City, NJ

May 2020 – Aug 2022

- Designed conceptual, logical, and physical data models for Santander's CECL and IFRS 9 credit risk analytics platform: defined entities, relationships, and attributes for consumer lending data including loan-level attributes, borrower characteristics, payment histories, and macroeconomic factor tables.
- Translated CECL regulatory business requirements into dimensional data structures; built star schema models for credit risk reporting, including fact tables for loan performance and dimension tables for borrower segments, product types, and time periods — maintaining 98% alignment with CECL data governance standards.
- Built and maintained ETL/ELT pipelines in SAS and SQL to extract, transform, and load large-scale consumer lending datasets from multiple relational source systems; implemented data quality checks, reconciliation workflows, and lineage documentation — reducing processing time by 10–15%.
- Created and maintained comprehensive data dictionaries and metadata documentation for CECL and IFRS 9 model data assets; ensured data consistency across allowance, capital planning, and loss forecasting data environments; participated in SOX control design for ALLL data governance processes.

Independent

Independent Consulting, Professional Development & Continued Education Oct 2019 – May 2020

- Provided independent data modeling and analytics consulting for financial institutions; deepened expertise in dimensional modeling, data warehouse design, and Snowflake data platform architecture; pursued continued education including MS Computer Science coursework with focus on database systems, data engineering, and cloud data platforms.

Huxley / Allegis Global Solutions (HSBC)

Data Modeler & Risk Analytics Specialist, Contract

New York, NY
Jun 2018 – Oct 2019

- Designed and maintained data models for HSBC's fraud risk analytics platform using SAS and SQL; defined entity relationships, data flows, and dimensional structures for transactional fraud datasets — maintaining 99% data quality compliance and enabling a 15% improvement in model efficacy through improved data structure design.
- Created and maintained data dictionaries and metadata documentation; collaborated with IT, DBAs, and model implementation teams to ensure data models accurately reflected business requirements and supported downstream analytics — achieving 95% documentation completeness.
- Optimized SQL query performance and data pipeline throughput; implemented proper indexing and schema design improvements that reduced monitoring activity time by 20%.

CRISIL / S&P Global

Data Modeler & Quantitative Analyst

New York, NY
Aug 2015 – Nov 2017

- Designed and maintained data models and SAS-based ETL pipelines for CCAR macroeconomic analytics platforms at Citi, Bank of America, and BNP Paribas; defined data entities, relationships, and dimensional structures for time series datasets — reducing data processing time by 20% and strengthening data governance for Capital Planning Working Groups.
- Created and maintained data dictionaries, source-to-target mappings, and metadata documentation; ensured data consistency across client analytical platforms — collaborating with data engineers, DBAs, and quantitative analysts to support reliable, scalable data architectures.

Kodukulla and Associates Inc.

Data Analyst & Research Analyst

Chicago, IL
Jan 2015 – May 2015

- Designed data models and analytical database structures for quantitative risk analysis using Monte Carlo Simulation and DCF modeling; created data dictionaries and maintained structured datasets supporting financial analysis — improving data organization and estimation accuracy by 10%.

EDUCATION

MS, Computer Science (Expected August 2027) Monroe University, New Rochelle, NY — focus: Database Systems, Data Engineering, Cloud Platforms

MS, Analytics — Harrisburg University of Science and Technology, Harrisburg, PA

MS, Finance — Risk Management & Corporate Finance — Illinois Tech, Stuart School of Business, Chicago, IL

MBA, Financial Management — Goldey-Beacom College, Wilmington, DE

CERTIFICATIONS

Apache PySpark by Example | End-to-End Data Engineering Project | Data Ingestion with Python | Bloomberg Essentials — Equity Essentials

WORK AUTHORIZATION

Legally authorized to work in the United States. No sponsorship required. H-1B cap-exempt with ~4 years remaining. Local to Jersey City, NJ — available for 5-day onsite immediately.